National University of Computer and Emerging Sciences

Lab Manual 06
Object Oriented Programming
(CL1004)

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Objectives:

After performing this lab, students shall be able to:

- ✓ Understand shallow copy and deep copy
- ✓ Create a copy constructor
- ✓ Pass objects as parameters to functions
- ✓ Use *this pointer

TASK:

Implement a class called **BiggerInt**. The BiggerInt class will have two data members:

- int* big_int_; // Pointer to the int array that holds the big integer
- int int_length_; // Variable to store the length of the big integer

While an integer is of 4 bytes in size with a range of -2,147,483,648 to 2,147,483,647. A big integer can store long integer numbers with no size limitation.

You have to implement the following:

1. Write a default constructor and initialize big_int_ to nullptr.
   - BiggerInt();

2. Write an overloaded constructor and perform deep copy.
   - BiggerInt (const int * obj, int size);

3. Write a copy constructor and perform deep copy. Print "Copy Constructor Called" and observer the scenarios where the copy constructor is called.
   - BiggerInt (const BiggerInt & obj);

4. Write a member function to make a deep copy of the big_int_ of the passed BiggerInt obj into the big_int_ of the object which called this function.
   - void assign(const BiggerInt & obj);

5. Write a member function which will overload the above assign function and performs the same operations but the argument passed to this function is a pointer integer array.
   - void assign(const int * big_int, int size);

6. Write a member function to append the big_int_ of the passed BiggerInt obj to the end of big_int_ of the object which called this function.
   - void append(const BiggerInt & obj);

7. Write a member function which will overload the above append function and performs the same operations but the argument passed to this function is a pointer integer array.
   - void append(const int* big_int, int size);

8. Write a member function to compare the big_int_ of BiggerInt obj with the big_int_ of the object which called this function. Return 0 for equal, 1 for less than and 2 for greater than.
    - int compareTo(const BiggerInt & obj);

9. Write a member function which overloads the above compareTo function and performs the same operations but the argument passed to this function is a pointer integer array.
    - int compareTo(const int* big_int, int size);

10. Write a member function to display the big_int_ on screen. If big_int_ is empty, print "No Value Assigned".
    - void display();

11. Write a destructor to deallocate any dynamically allocated memory.
    - ~ BiggerInt();

12. Write a suitable main() function to test all the functions of the BiggerInt class.

Note:

- Deallocate all dynamically allocated memory.
- Do not use any string class built-in functions except for strlen(), if required.
- Follow all the code indentation, naming conventions and code commenting guidelines.